Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2001-I-Q05] | Part A (Q1-Q5)

An equilateral triangle lies on $x^2 + 4y^2 = 4$, has vertex $(0, 1)$, and has an altitude on the y-axis. Its squared side length is m/n reduced. Find $m + n$.

等边三角形内接于 $x^2 + 4y^2 = 4$，一个顶点为 $(0, 1)$，一条高位于 y 轴。边长平方约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-I-Q09] | Part B (Q6-Q10)

A rhombus has vertices on $x^2/20 - y^2/24 = 1$ and its diagonals meet at the origin. Find the greatest real number strictly smaller than BD^2 for every such rhombus $ABCD$.

菱形 $ABCD$ 的顶点在双曲线 $x^2/20 - y^2/24 = 1$ 上，两对角线交于原点。求对所有这种菱形均严格小于 BD^2 的最大实数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2025-I-Q09] | Part B (Q6-Q10)

Rotate $y = x^2 - 4$ counterclockwise by 60° about the origin. The unique fourth-quadrant intersection of the original curve and its image has $y = (a - \sqrt{b})/c$, with positive integers and coprime a, c. Find $a + b + c$.

将抛物线 $y = x^2 - 4$ 绕原点逆时针旋转 60°。原曲线与像曲线在第四象限的唯一交点，其纵坐标为 $(a - \sqrt{b})/c$，其中为正整数且 a, c 互质，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2026-II-Q06] | Part B (Q6-Q10)

Sum all radii r for which a circle centered at $(4, 39)$ is tangent at some point to the parabola $2y = x^2 - 8x + 12$.

求所有使圆心 $(4, 39)$ 的圆与抛物线 $2y = x^2 - 8x + 12$ 在至少一点相切的半径 r 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2005-II-Q15] | Part C (Q11-Q15)

Circles are given by $x^2+y^2+10x-24y-87=0$ and $x^2+y^2-10x-24y+153=0$. Find the least positive slope a of a line $y=ax$ containing the center of a circle internally tangent to the first and externally tangent to the second. If $a^2=p/q$ is reduced, find $p+q$.

两圆方程为 $x^2+y^2+10x-24y-87=0$、$x^2+y^2-10x-24y+153=0$。直线 $y=ax$ 包含某圆圆心，该圆与第一圆内切、第二圆外切。求最小正斜率 a，若 $a^2=p/q$ 最简，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2021-I-Q15] | Part C (Q11-Q15)

For positive integers k, consider $y = x^2 - k$ and $x = 2(y-20)^2 - k$. Among k for which they have four distinct intersections lying on a circle of radius at most 21, add the least and greatest values.

对正整数 k，考虑 $y = x^2 - k$ 与 $x = 2(y-20)^2 - k$。要求两曲线有四个不同交点，且这些点位于半径不超过 21 的同一圆上。求符合条件的最小与最大 k 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2022-II-Q12] | Part C (Q11-Q15)

For real x, y and parameters $a > 4, b > 1$, suppose $x^2/a^2 + y^2/(a^2 - 16) = 1$ and $(x - 20)^2/(b^2 - 1) + (y - 11)^2/b^2 = 1$. Determine the minimum of $a + b$.

实数 x, y 及参数 $a > 4, b > 1$ 满足 $x^2/a^2 + y^2/(a^2 - 16) = 1$，$(x - 20)^2/(b^2 - 1) + (y - 11)^2/b^2 = 1$。求 $a + b$ 的最小值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

[AIME-2024-II-Q12] | Part C (Q11-Q15)

Let $O=(0,0), A=(1/2,0), B=(0,\sqrt{3}/2)$. Among all unit segments joining the positive coordinate axes, there is exactly one interior point C of AB lying on no other such segment. Write $OC^2=p/q$ in lowest terms and find $p+q$.

令 $O=(0,0), A=(1/2,0), B=(0,\sqrt{3}/2)$。考虑所有连接两正坐标轴且长为 1 的线段。在 AB 内有唯一一点 C 不属于其他这种线段。若 $OC^2=p/q$ 为最简分数，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程